

Smart Attendance Monitoring System Based on RFID and Arduino Using IoT Connectivity

**Written to fulfill the 1st task of
Information Systems Architecture and Technology**



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2025

PREFACE

We would like to express our gratitude to God Almighty for the blessings, health, and guidance that have enabled us to complete this academic work titled “Smart Attendance Monitoring System Based on RFID and Arduino Using IoT Connectivity.” This project is the result of our collective effort and aims to provide a clear analysis of how RFID, Arduino, and IoT technologies can be integrated to build an efficient and reliable attendance monitoring system.

During the preparation of this work, we gained meaningful insights into the role of microcontrollers, identification technologies, and cloud-based communication in modern automation systems. This study also allowed us to explore how automated attendance systems can help organizations reduce manual errors, improve accuracy, and provide real-time data monitoring for better administrative decision-making.

This work is driven by the rapid development of digital automation and the increasing need for smart solutions in corporate and industrial environments. We hope that this paper can serve as a useful academic reference for students, developers, and professionals who are interested in embedded systems, IoT applications, and automation technologies.

We would also like to express our sincere appreciation to our instructors, classmates, and everyone who supported us throughout the completion of this project. We acknowledge that this work is not without limitations, and we welcome constructive feedback from readers to help us improve and refine it further in the future.

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CHAPTER I

INTRODUCTION

1.1 Background

The rapid development of Internet of Things (IoT) technology has transformed the way institutions manage operational processes, particularly in attendance monitoring. Traditional attendance methods, such as manual signing or punch-card systems, often suffer from inaccuracies, data manipulation, and inefficiency, especially in environments involving large groups of people (Shrivastava et al., 2023). These limitations have encouraged organizations to adopt automated attendance solutions that provide accuracy, transparency, and real-time data processing.

Radio Frequency Identification (RFID) has become one of the most widely used technologies for automation due to its reliability and contactless identification capabilities. RFID systems allow users to authenticate themselves simply by placing an RFID tag near a reader, making the process faster, more secure, and less prone to human error compared to barcode or manual-based systems (Bharathy et al., 2021). Studies show that RFID-based attendance systems significantly reduce time delays and eliminate manual inconsistencies in data collection (IEEE, 2019).

The integration of RFID with microcontroller platforms, such as Arduino, further enhances system flexibility. Arduino provides an open-source, low-cost, and highly customizable environment, making it suitable for building intelligent attendance systems tailored to specific institutional needs. In recent research, Arduino-based attendance systems paired with Wi-Fi modules such as ESP8266 successfully transmitted attendance records to cloud-based platforms, demonstrating high efficiency in data handling and remote accessibility (Shrivastava et al., 2023).

IoT connectivity plays a crucial role in modernizing attendance systems. By enabling continuous data transmission to cloud servers or centralized dashboards, IoT-based attendance systems allow administrators to monitor attendance in real time, receive instant alerts, and generate automated reports. A recent systematic literature review highlighted that IoT-based attendance systems improve data transparency, reduce administrative workload, and provide better scalability for institutions managing large populations (Ishaq & Bibi, 2023).

Further supporting this development, studies presented in IEEE conferences have demonstrated that IoT-based RFID attendance architectures improve data security by integrating authentication protocols, encrypted communication, and database verification (IEEE, 2020; IEEE, 2024). These findings emphasize the effectiveness of combining RFID, Arduino, and IoT infrastructure to create a robust and intelligent attendance monitoring system.

Given the increasing demand for automation and accurate data tracking, the **Smart Attendance Monitoring System Based on RFID and Arduino Using IoT Connectivity** offers a modern, efficient, and scalable approach to attendance management. Its ability to provide contactless identification, real-time monitoring, and cloud-based data storage makes it an ideal solution for schools, universities, workplaces, and public institutions that require reliable and automated attendance tracking (Bharathy et al., 20

1.2 Writing Objective

To explain the design, implementation, and evaluation of an RFID-, Arduino-, and IoT-based Smart Attendance System, and how it improves accuracy and efficiency compared to manual methods.

1.3 Problem Domain

Manual attendance is prone to fraud, slow, and often inaccurate. This study focuses on how RFID, Arduino, and IoT provide a faster, real-time, and more reliable solution.

1.4 Writing Methodology

The report uses literature review, device analysis, and technical examination based on journals, official documentation, and online references to explain system concepts and effectiveness.

1.5 Writing Framework

The structure of this ISAT is organized as follows:

CHAPTER I: INTRODUCTION

Contains the background of the problem, objectives, and the importance of IoT-based automated attendance systems.

CHAPTER II: THEORETICAL BACKGROUND

Explains the theories related to RFID, Arduino, IoT, and digital attendance systems.

CHAPTER III: SYSTEM ANALYSIS AND DESIGN

Discusses the overall structure of the Smart Attendance Monitoring System, including an overview of how the system operates, the main objectives of the automated

attendance solution, and the integration workflow between RFID, Arduino, and IoT connectivity. This chapter explains how each component interacts within the system, how data flows from the RFID tag to the server, and how IoT enables real-time monitoring and centralized attendance management.

CHAPTER IV: CONCLUSION AND SUGGESTION

Summarizes the findings and provides suggestions for future system development.

CHAPTER II

BASIC THEORY

2.1 Definition of IoT

The internet of things, or IoT, is a network of interrelated devices that connect and exchange data with other IoT devices and the cloud. IoT devices are typically embedded with technology, such as sensors and software, and can include mechanical and digital machines and consumer objects.

These devices encompass everything from everyday household items to complex industrial tools. Increasingly, organizations in various industries are using IoT to operate more efficiently, deliver enhanced customer service, improve decision-making and increase the value of the business.

With IoT, data is transferable over a network without requiring human-to-human or human-to-computer interactions.

A thing in the internet of things can be a person with a heart monitor implant, a farm animal with a biochip transponder, an automobile that has built-in sensors to alert the driver when tire pressure is low, or any other natural or human-made object that can be assigned an Internet Protocol address and can transfer data over a network.

2.2 Definition of RFID

RFID is a technology that uses electromagnetic fields to automatically identify and track tags attached to objects. An RFID system typically includes a tiny radio transponder (tag), an RFID reader (which both transmits and receives), and an antenna. When the reader sends out a pulse of electromagnetic energy, the tag is activated and transmits its stored digital data (usually a unique ID number) back to the reader.

There are passive tags, which draw their power from the reader's signal, and active tags, which have their own power source and can be read from much further away.

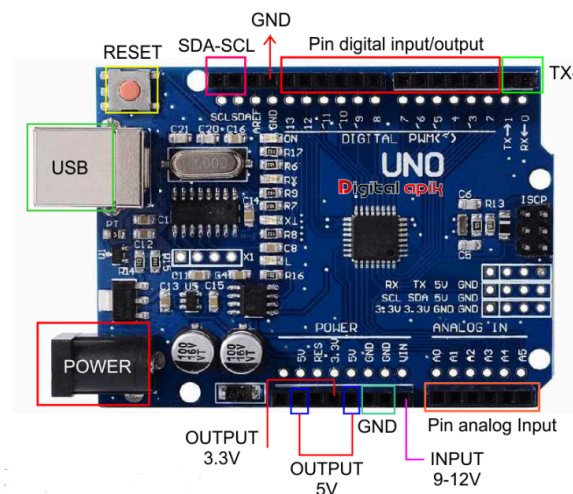
2.3 Definition of Arduino

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing

something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments.



2.1 Arduino, REFF: <https://vication.oemahwebsite.com/pengertian-arduino>

USB Connector

1. Upload program from the computer to the board.
2. Provide electrical power to the Arduino board and input/output components.

Reset Button

1. Reset the Arduino board, so the program will start from the beginning. Reset only reloads, it does not delete the program already on the Arduino.

Digital I/O Pin

1. Consists of 14 pins, which are pins 0-13.
2. Functions to place input/output components.

Analog Input Pin

1. Consists of 6 Analog input pins, which are pins A0-A5.
2. Functions to read the voltage from Analog sensors, such as temperature sensors.

Power Jack

1. Jack for a battery or external power source. If a battery is already being used, there is no need to use a USB cable connected to the computer.

VCC Pin / Positive (+)

1. Provides positive (+) electrical supply with a voltage of 5V.
- GND Pin / Negative (-) Provides negative (-) electrical supply

CHAPTER III

SYSTEM ANALYSIS AND DESIGN

3.1 Overview of the Smart Attendance Monitoring System

The Smart Attendance Monitoring System based on RFID and Arduino with IoT connectivity is an automated platform designed to record, store, and monitor attendance data digitally in real time. It replaces manual paper-based processes by using RFID cards, a microcontroller (Arduino), and networked databases or cloud services to improve accuracy, efficiency, and transparency.

A smart attendance system uses RFID tags or cards assigned to each user, an RFID reader connected to an Arduino-based controller, and a backend database accessible via IoT communication. When a user taps the RFID card on the reader, the controller identifies the unique ID, validates it, and automatically logs the attendance event with time and status into a local or cloud database. This architecture minimizes manual intervention and supports centralized management of attendance records through web or mobile interfaces.

3.2 Main objectives of the system

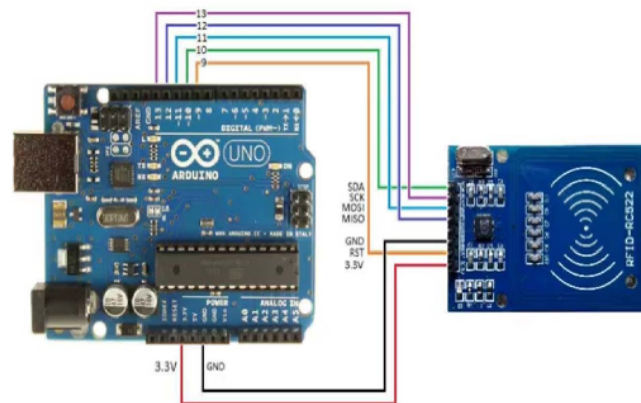
The primary objective is to automate the attendance process so that presence is recorded instantly by scanning RFID cards rather than filling paper sheets or calling names. This automation reduces processing time, lowers administrative workload, and supports scalable deployment in schools, universities, and workplaces.

A second objective is real-time monitoring, where attendance data are transmitted via IoT connectivity to a server or cloud platform, enabling stakeholders to view up-to-date presence information through dashboards, notifications, or reports. Some implementations add real-time alerts using GSM, email, or messaging APIs to inform parents, staff, or administrators about absences or late arrivals.

The system also aims to reduce human error and fraud by eliminating manual entry, preventing proxy attendance, and ensuring that only registered RFID tags can be used to mark presence. Automatic data capture and digital storage improve data integrity, make it easier to audit attendance records, and reduce the risk of lost or damaged paper documents.

3.3 Integration of RFID, Arduino, and IoT

In this system, RFID technology provides the identification layer, where each tag contains a unique identifier that is read contactlessly by an RFID reader module such as MFRC522. The reader sends the tag's unique ID to an Arduino-based microcontroller (e.g., Arduino Uno, Nano, or Wi-Fi-enabled variants) via SPI or UART communication. To begin with implementation, Arduino UNO is connected to MFRC522. Once the connection is set up, the library for The RFID module is downloaded and configured to the Arduino UNO. Following is the Pin Configuration for the Implementation Setup.



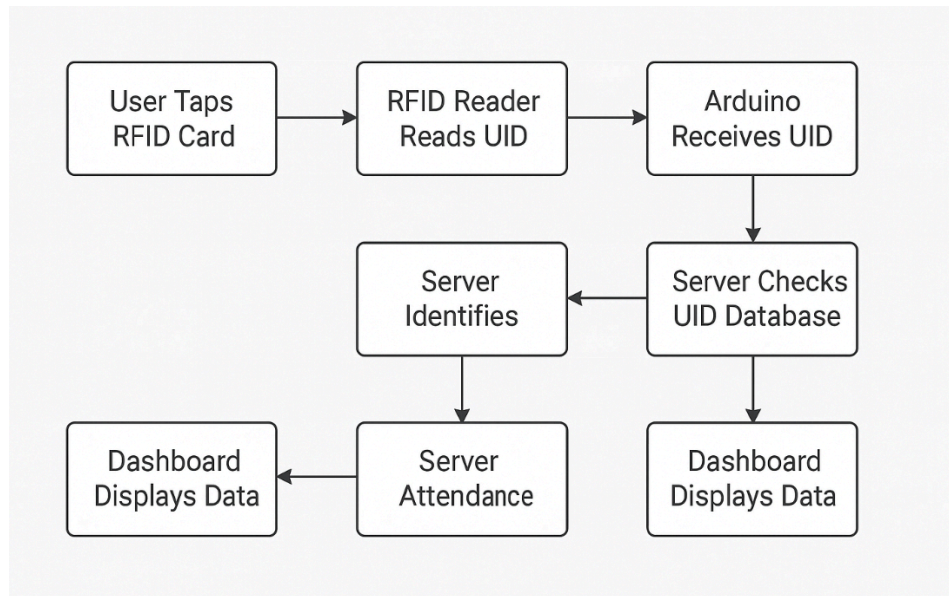
3.1 Rangkaian RFID Reader - Arduino, REFF:

<https://medium.com/@ifan.ramadhanzaki/rfid-6892a2aa6ce8>

The Arduino acts as the core processing unit that controls the RFID reader, validates card IDs, manages local peripherals (LCD, buzzer, keypad), timestamps events, and formats the data for transmission. Using built-in or external communication modules (Wi-Fi, Ethernet, GSM), the Arduino forwards attendance records to a backend system, which can be a local server or a cloud database accessible over the Internet.

IoT connectivity links the embedded attendance terminal with remote services, enabling features such as cloud storage, web dashboards, analytics, and mobile access. In many designs, the server exposes RESTful APIs or web applications that store RFID-based logs in databases like MySQL and provide visualization, reporting, and real-time status monitoring for administrators and teachers.

3.4 Integration Workflow



smart attendance system

In this Smart Attendance system, each RFID card has a different UID for each user.

When the card is tapped, the server will check whether the UID belongs to the category:

Student	Lecturer	employee
UID: 0001abcc	UID: 1001abcc	UID: 1111abcc

1. After tap, Arduino Receives UID

Arduino receives the UID from the reader.

Arduino immediately wraps the data into JSON, example student:

```

1 json
2 {
3   "uid": "0001F12C",
4   "timestamp": "2025-12-03 08:10:55",
5   "device_id": "GATE01"
6 }
7

```

example student

2. Arduino Sending Data to the Server (IoT)

Data is sent via:

- WiFi
- or MQTT
- or HTTP POST

In essence: from the device to the server/cloud.

3. Server Checks UID in Database

The server immediately searches for the UID in the user table.

From here, the server knows:

UID	Name	Role
0001F12C	Muhammad Akasyah	Student
1001AB6Y	mr hasan	lecturer
1111KL73	mr agung	employee

4. The Server Determines Actions Based on Role

student	lecturer	employee
• Class attendance • Data sent to the class teacher's dashboard • Can be used for daily attendance records	• The lecturer is present to teach. • Activate attendance status for that day's teaching schedule. • This may trigger the opening of the class attendance page on the teacher's dashboard.	• Daily attendance (arrival or departure) • Recorded for HR/administrative purposes • Not related to classes

5. Dashboard Displays Data in Real Time

The data appears immediately in the application or web dashboard:

- Teachers can see student attendance.
- Academics can see lecturer attendance.
- Administrators can see staff attendance.

Real-time, because the IoT platform updates immediately as soon as the data comes in.

CHAPTER IV

CONCLUSION AND SUGGESTION

4.1 Conclusion

The Smart Attendance Monitoring System is an automated platform that digitally records and monitors attendance data in real-time. It successfully replaces manual processes by integrating RFID for unique user identification, an Arduino microcontroller as the core processing unit, and IoT connectivity to transmit records to a cloud database. This integration ensures instant presence recording, significantly reduces administrative workload, and improves data integrity by eliminating manual error and preventing proxy attendance.

4.2 Suggestion

To ensure the system's long-term reliability and robustness, it is suggested to prioritize implementing robust local data buffering on the Arduino to prevent data loss during inevitable internet or server outages. Furthermore, administrators should enhance the security protocols for data transmission to the cloud database and optimize the web interface to provide more advanced, intuitive analytics and reporting features for stakeholders.

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